

CampusFlow™ Academic Operating System

Full-Stack Enterprise Technical Architecture, Problem Analysis & Optimizations

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1. Executive Summary & Product Vision

CampusFlow is a modern, high-concurrency academic operating platform designed to replace disorganized, unsecured messaging apps (WhatsApp, Telegram) with an official, institutional-grade campus network. It unifies real-time semester discussion channels, curriculum-organized syllabus storage, contextual Retrieval-Augmented Generation (RAG) academic AI, and strict role verification into a single ultra-responsive web application.

2. Critical Engineering & Business Problems Solved

Problem Encountered	Root Cause / Legacy Deficiency	Engineered Solution in CampusFlow
1. Academic Information Chaos & Inter-Semester Leakage	Students across Sem 1 to 8 mixed into identical channels and accessed unauthorized notes without boundaries.	Strict Multi-Layer Semester Isolation: Enforced across backend SQL queries, WebSocket rooms, course note views, and AI assistant context. Sem 1 students can only view Sem 1 resources.
2. Unusable S3 Upload Latency (2–3 sec wait)	AWS S3 bucket located in Sydney (ap-southeast-2) incurred 2000–3000ms round-trip latency, blocking the API thread during photo uploads.	Local Persistent Cache + Async S3 Task: Immediate local disk write (<10ms) returns instant 200 OK to the client; AWS S3 upload executed asynchronously in background via <code>asyncio.create_task</code> .
3. Profile Photo Disappearing / Resetting on Login	Database stored raw internal S3 keys (avatars/...). Login API returned this raw string, leading the browser to resolve relative URLs with 404 errors.	Unified Endpoint Normalization: All auth endpoints now format avatars into <code>/api/profile/avatar/{id}/</code> with persistent disk fallback and client-side cache busting (<code>?t=...</code>).
4. AI Hallucinations on Course Exams	Generic LLMs lack institution-specific textbooks, syllabus guidelines, and teacher slides, yielding inaccurate answers.	Syllabus-Grounded RAG Pipeline: Uploaded lecture PDFs are chunked (800 chars), embedded via Sentence-Transformers, and stored in vector memory for sub-second grounded question answering.
5. Intrusive Technical Developer Jargon	User-facing UI contained developer buzzwords like 'Live WebSocket', 'Update S3 Photo', 'Gmail OTP', and 'Uploading to S3...'.	Production Polish: Completely scrubbed all infrastructure terms; replaced with intuitive terms like 'Online', 'Update Photo', and 'Continue with Google'.

3. Full-Stack Technology Matrix

Tier / Domain	Selected Technology	Architectural Role & Value Proposition
Frontend Framework	Next.js 14 (App Router) + React 18 + TypeScript	Server-Side Rendering (SSR) & Static Page Generation for 13 optimized routes, strict type safety, zero layout shifts.
Styling & UX	Tailwind CSS + Lucide Icons	Modern White + Emerald Green theme (#059669), fluid responsive mobile-drawer layouts, custom animated micro-interactions.
Client State & Caching	TanStack React Query v5	Automatic background refetching, declarative query keys (scoped by semester), optimistic cache updates, zero redundant network calls.
Backend API	FastAPI (Python 3.12 ASGI)	High-concurrency async non-blocking request handling, automated OpenAPI documentation, sub-10ms JSON response speed.
Asynchronous ORM	SQLAlchemy 2.0 (AsyncSession) + SQLite / PostgreSQL	Relationship eager-loading (selectinload) eliminating N+1 query overheads across subjects, users, notices, and messages.
Real-Time Engine	FastAPI WebSockets + ConnectionPool	Full-duplex real-time messaging, client presence detection, broadcasting WhatsApp-style messages with DP avatar payloads.
In-Memory Store & 2FA	Redis 7 Alpine	TTL-backed (5-minute auto-expiry) one-time login verification codes (OTP), session caching, and rate limiting.
Object Storage	Amazon Web Services (AWS S3) + Boto3	Durable multi-part cloud storage for syllabus PDFs, presentation decks, lecture images, and avatars with presigned URL streaming.
Academic AI & Vector Search	Sentence-Transformers + Groq / DeepSeek LPU	Local chunk embedding vector store enabling sub-second Retrieval-Augmented Generation (RAG) grounded in syllabus PDFs.
Video Conferencing	Jitsi Meet Web API	1-click instant class video conferencing rooms isolated dynamically by Semester and Subject Code.
Infrastructure & DevOps	Docker + Docker Compose + Nginx	Containerized multi-service orchestration (Backend, Frontend, Redis, Nginx reverse proxy) deployed on AWS EC2.

4. Key Feature Implementations & Technical Deep-Dive

A. Real-Time Classroom Channels (College 'WhatsApp')

- **Full-Duplex WebSockets:** Room-level subscription pools allow students and professors to converse in real-time without polling.
- **WhatsApp-Style Circular Avatars:** Every message broadcasts sender initials or custom avatar DP from `/api/profile/avatar/{id}/`.
- **Presence Dot & Status:** Live connection states display clear indicators: *Online*, *Connecting...*, or *Offline*.
- **1-Click Class Video Meet:** Integrated Jitsi Meet launches high-definition video classrooms with zero registration required.

B. Strict Multi-Tier Semester Isolation

- **Chat Sandboxing:** Semester 1 students only receive and broadcast messages in Semester 1 subject channels; cross-semester joins are blocked by HTTP 403 / WebSocket 4003.
- **Study Notes Sandboxing:** The student portal automatically locks queries to the student's enrolled semester (e.g. Semester 1 Official Notes). The 8-semester switch tabs are completely removed for students.
- **AI Assistant Scope:** The Course Context dropdown under /ai-assistant filters available courses strictly to the student's active semester.

C. NexusAI Syllabus RAG Assistant

- **Automated PDF Ingestion:** When professors upload course notes, the backend automatically extracts text and creates 800-character chunks with 100-character overlaps.
- **Vector Embeddings:** Chunks are embedded via Sentence-Transformers into a persistent vector index.
- **Sub-Second Retrieval:** Student questions trigger cosine similarity vector queries across their syllabus documents, sending top chunks as context to the LLM for hallucination-free answers.

D. Digital Campus ID Card & Profile Management

- **Interactive Academic Badge:** Displays verified Student/Faculty badge, dynamic USN identifier, and simulated security barcode.
- **Instant Photo Updates:** Uploaded images update in milliseconds with immediate client-side preview, cache-busting timestamps, and background cloud sync.

5. Systematic Performance & Latency Optimizations

Key Optimization Strategies Implemented:

1. **Asynchronous I/O Offloading:** Separated time-critical HTTP responses from slow cloud writes. Avatar uploads save to local persistent volume (/app/data/avatars/) in <10ms, while AWS S3 upload is dispatched in an asynchronous task (asyncio.create_task).
2. **Sub-Millisecond Disk Serving:** get_user_avatar inspects the local disk first using pattern globbing. If present, it returns an immediate FileResponse (0–2ms) instead of streaming over WAN from S3.
3. **S3 First-Hit Auto-Caching:** When an avatar or document is first requested from S3, the server streams bytes to the client and simultaneously persists a local disk copy for all subsequent requests.
4. **Browser Cache Invalidation:** Cleaned image caching bugs by appending deterministic cache-buster timestamps (?t=\${Date.now()}) upon upload, ensuring zero stale images.
5. **Client-Side Query Deduplication:** TanStack React Query handles server-state caching, eliminating redundant network calls across tab switches and route transitions.
6. **Production Bundle Minimization:** Next.js production compilation achieves optimized bundle sizes (average first-load JS ~120-140 kB across 13 routes).

6. Production Deployment Topology (AWS EC2)

CampusFlow runs in a multi-container isolated Docker network orchestrated via Docker Compose:

- **Nginx Container (Port 80/443):** Handles incoming web traffic, TLS termination, reverse proxy routing, and WebSocket upgrades.
- **FastAPI Backend (Port 8000):** High-concurrency Python ASGI worker with mounted persistent volume ./backend/data:/app/data.
- **Next.js Frontend (Port 3000):** Optimized standalone Node.js production server with dynamic runtime environment variables.
- **Redis Container (Port 6379):** In-memory cache for fast 2FA OTP codes and session invalidation.

System Status: Production Ready	Target Domain: campus-flow.duckdns.org	Build Validation: 0 Errors (13 Routes)
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